

Configuring the Windows Defender Firewall (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 04

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Time on Task:

2 hours, 21 minutes

Progress:

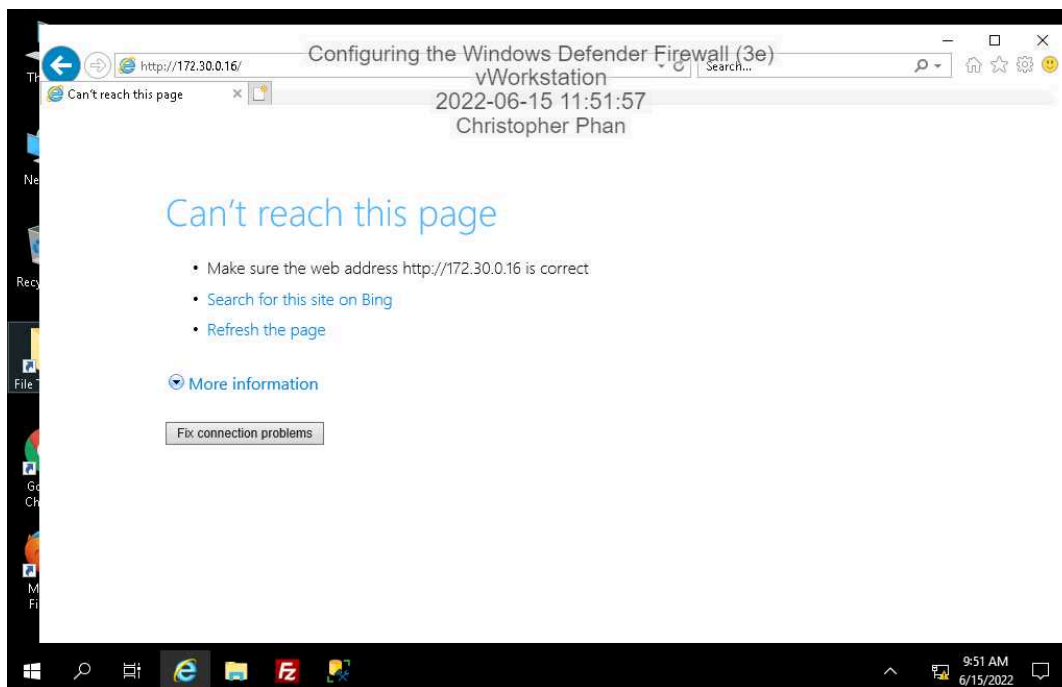
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Report Generated: Wednesday, June 15, 2022 at 1:37 PM

Section 1: Hands-On Demonstration

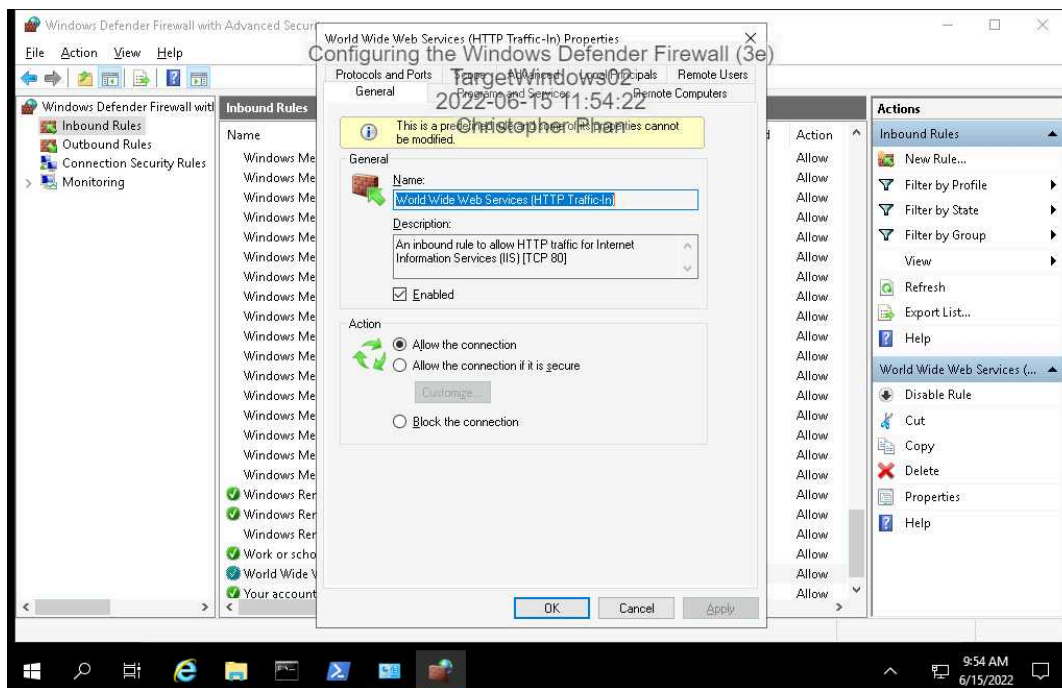
Part 1: Test Access to a Windows IIS Server Using a Web Browser

3. Make a screen capture showing the connection error message in the browser window.



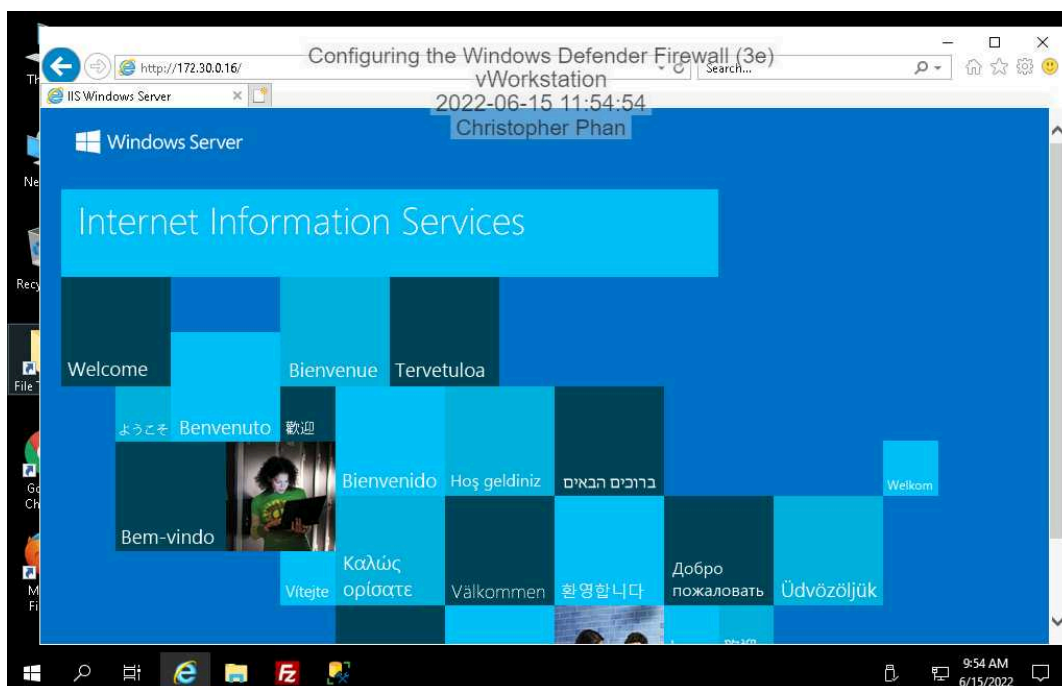
Part 2: Enable a Windows Defender Firewall Rule to Allow HTTP Access

12. Make a screen capture showing the **World Wide Web Services (HTTP Traffic-In)** Properties dialog box and the **Enabled** checkbox.



Part 3: Confirm the Windows Defender Firewall Configuration Allows HTTP

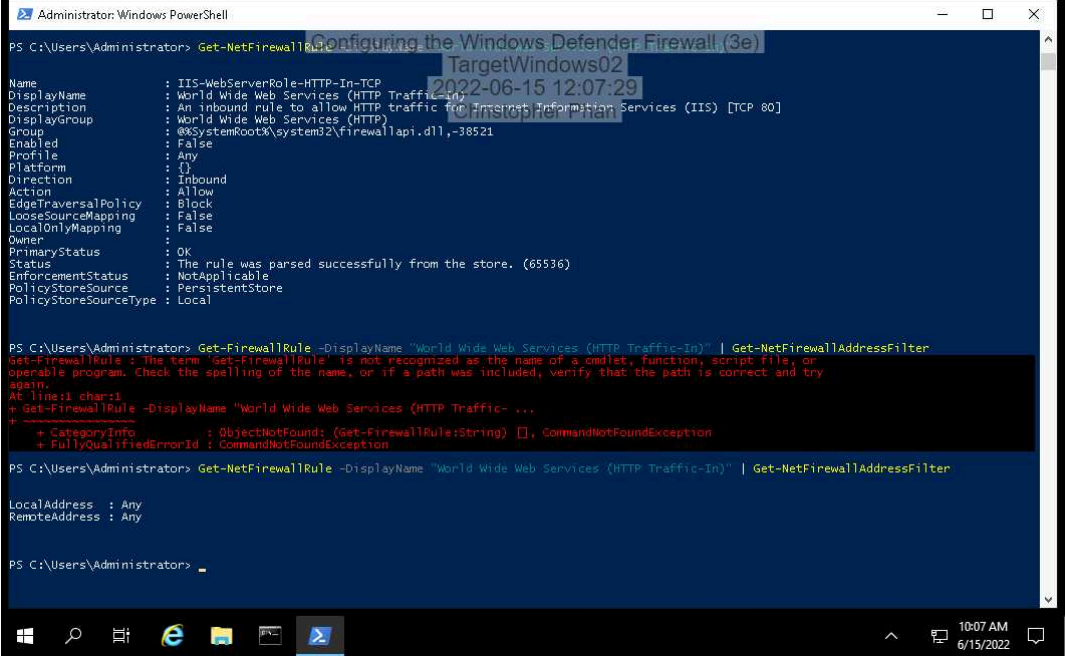
4. Make a screen capture showing the IIS server home page in the browser window.



Section 2: Applied Learning

Part 1: Test Access to Windows IIS Server Using PowerShell

6. Make a screen capture showing the results of the **Get-NetFirewallRule** commands.



```
Administrator: Windows PowerShell

PS C:\Users\Administrator> Get-NetFirewallRule -Name "World Wide Web Services (HTTP Traffic-In)"

Name                : IIS-WebServerRole-HTTP-In-TCP
DisplayName          : World Wide Web Services (HTTP Traffic-In)
Description          : An inbound rule to allow HTTP traffic for Internet Information Services (IIS) [TCP 80]
DisplayGroup         : World Wide Web Services (HTTP)
Group                : @SystemRoot\system32\Firewallapi.dll,-38521
Enabled              : False
Profile              : Any
Platform             : {}
Direction            : Inbound
Action               : Allow
EdgeTraversalPolicy   : Block
LooseSourceMapping    : False
LocalOnlyMapping      : False
Owner                :
PrimaryStatus         : OK
Status               : The rule was parsed successfully from the store. (65536)
EnforcementStatus     : NotApplicable
PolicyStoreSource     : PersistentStore
PolicyStoreSourceType : Local

PS C:\Users\Administrator> Get-NetFirewallRule -DisplayName "World Wide Web Services (HTTP Traffic-In)" | Get-NetFirewallAddressFilter

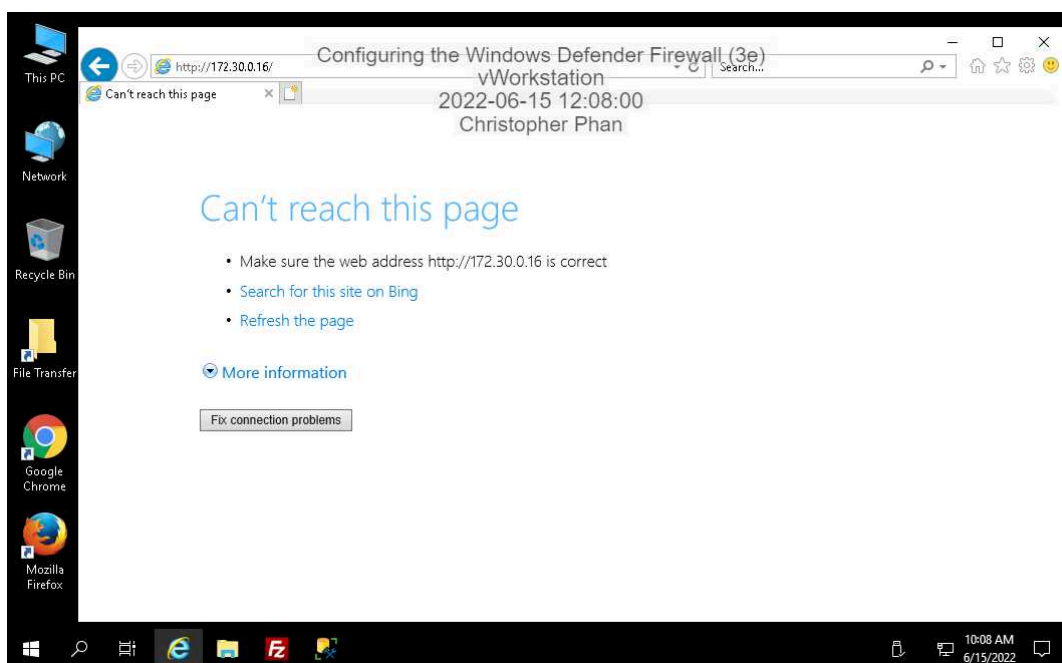
Get-NetFirewallRule : The term 'Get-NetFirewallRule' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ Get-NetFirewallRule -DisplayName "World Wide Web Services (HTTP Traffic-In)" | Get-NetFirewallAddressFilter
+ ~~~~~
    + CategoryInfo          : ObjectNotFound: (Get-NetFirewallRule:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Administrator> Get-NetFirewallRule -DisplayName "World Wide Web Services (HTTP Traffic-In)" | Get-NetFirewallAddressFilter

LocalAddress : Any
RemoteAddress : Any

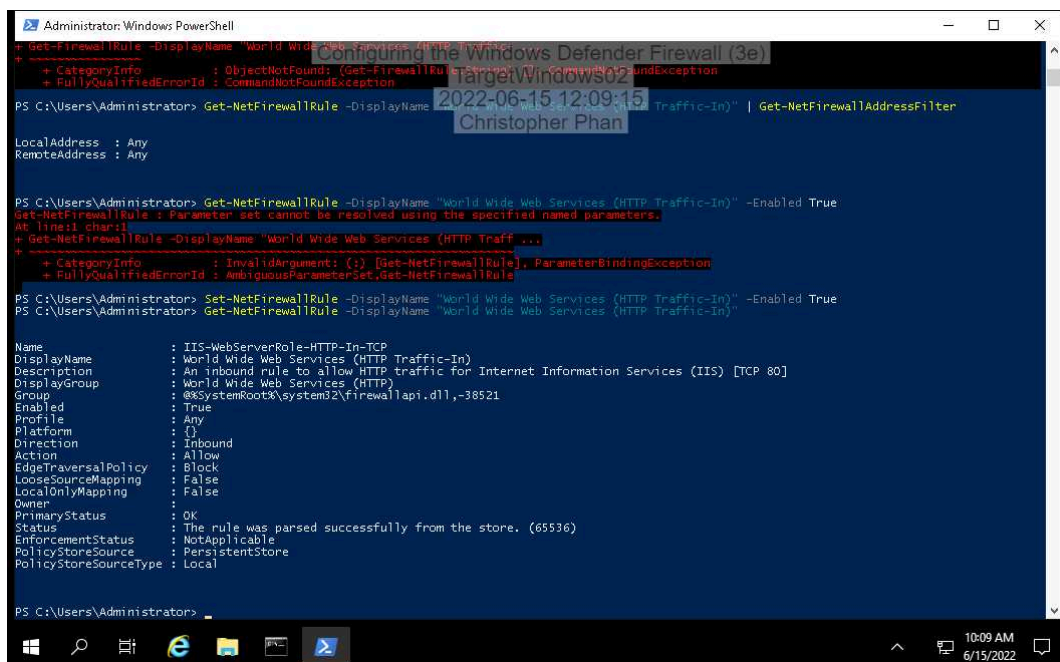
PS C:\Users\Administrator>
```

9. Make a screen capture showing the connection error message in the browser window.



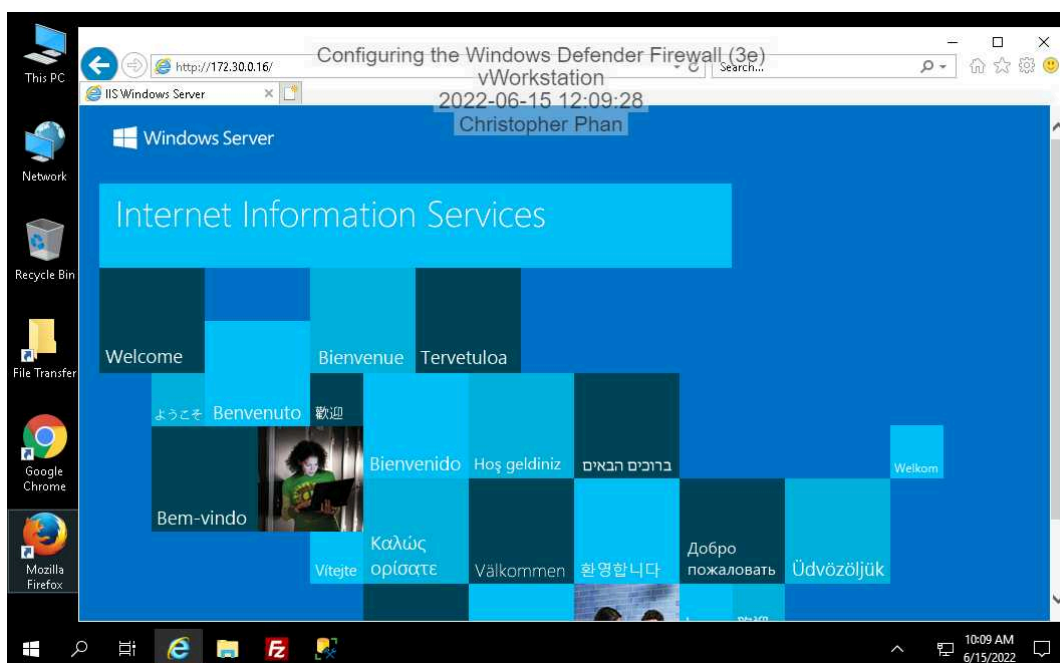
Part 2: Enable a Windows Defender Firewall Rule to Allow HTTP Access

4. Make a screen capture showing the updated status of the World Wide Web Services rule.



Part 3: Confirm the Windows Defender Firewall Configuration Allows HTTP

3. Make a screen capture showing the IIS server home page in the browser window.



Section 3: Challenge and Analysis

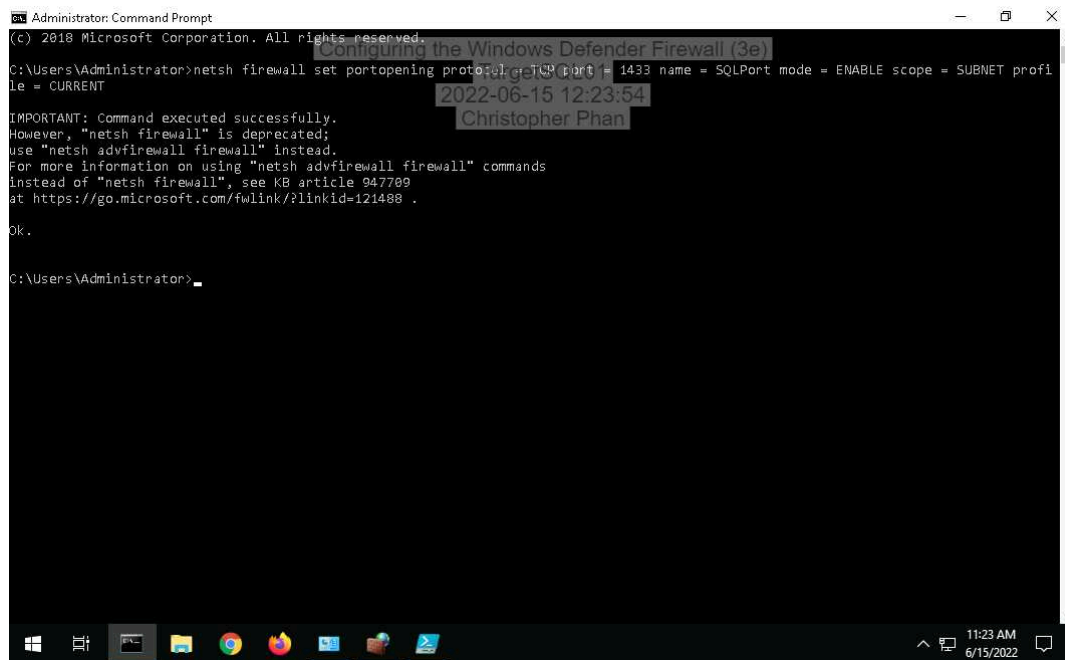
Part 1: Research Firewall Rule Requirements

Using the Internet, **research** and **identify** the protocol and port necessary for a Microsoft SQL Server database connection.

The protocol needed is a TCP protocol. The ports needed are TCP port 1433 and UDP port 1434, making these open allow for a Microsoft SQL Server database connection.

Part 2: Configure the Firewall

Make a screen capture showing your firewall rule on TargetSQL01.



```
Administrator: Command Prompt
(c) 2018 Microsoft Corporation. All rights reserved.
C:\Users\Administrator>netsh firewall set portopening protocol=TCP port=1433 name = SQLPort mode = ENABLE scope = SUBNET profile = CURRENT
IMPORTANT: Command executed successfully.
However, "netsh firewall" is deprecated;
use "netsh advfirewall firewall" instead.
For more information on using "netsh advfirewall firewall" commands
instead of "netsh firewall", see KB article 947789
at https://go.microsoft.com/fwlink/?linkid=121488 .
OK.
C:\Users\Administrator>
```

Part 3: Validate Your Firewall Rule

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Make a screen capture showing the results of your SQL query.

SQLQuery5.sql - 172.30.0.17.lab (bross (51)) - Microsoft SQL Server Management Studio (Administrator)

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2022-06-15 12:35:54

Christopher Phan

SELECT * FROM countries;

Results

	Country	Region	Population	Area_sq_mi	Pop_Density_per_sq_mi	Coastline_coast_area_ratio	Net_migration
1	Afghanistan	ASIA (EX. NEAR EAST)	31056997	647500	480	0	23.06
2	Albania	EASTERN EUROPE	3581655	28748	1246	126	-4.93
3	Algeria	NORTHERN AFRICA	32930091	2381740	138	4	-0.39
4	American Samoa	OCEANIA	57794	199	2904	5829	-20.71
5	Andorra	WESTERN EUROPE	71201	468	1521	0	6.6
6	Angola	SUB-SAHARAN AFRICA	12127071	1246700	97	13	0
7	Anguilla	LATIN AMER. & CARIB	13477	102	1321	5980	10.76
8	Antigua & Barbuda	LATIN AMER. & CARIB	69108	443	1560	3454	-6.15
9	Argentina	LATIN AMER. & CARIB	39921833	2766890	144	18	0.61
10	Armenia	C.W. OF IND. STATES	2976372	29800	999	0	-6.47
11	Aruba	LATIN AMER. & CARIB	71891	193	3725	3549	0
12	Australia	OCEANIA	20764082	7686950	26	24	2.98

Query executed successfully. 172.30.0.17 (14.0 RTM) | bross (51) | lab | 00:00:00 | 227 rows